

Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's Disease

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Project Summary

This project develops a closed-loop deep learning system for personalized 40 Hz auditory entrainment therapy in Alzheimer's disease. Current clinical protocols deliver stimulation on rigid fixed schedules, ignoring individual variability in neural responses and intra-session habituation. I analyzed EEG recordings from 35 elderly subjects (OpenNeuro ds005048) and discovered that dropping 61 spectral features in favor of 12 PAC-derived and stimulation context features raised forecasting accuracy from test $R^2 = 0.121$ (73 features) to 0.606 (5-seed mean, std = 0.032), driven by removing subject-specific EEG characteristics from model inputs. The predictive controller matched stimulation to patient need 72.1% versus 64.5% for reactive control ($p < 0.001$, Hedges' $g = 1.31$), targeted 82.6% of low-coupling windows versus 51.7%, and reached 91% of the theoretical oracle. All 35 subjects benefited. A caregiver-facing web application is deployed on Hugging Face Spaces to demonstrate the concept.

Introduction

Research Question

Can deep learning models trained on EEG-derived features forecast theta–gamma phase-amplitude coupling (PAC) dynamics 5–10 seconds into the future, and does integrating such forecasts into a closed-loop controller produce measurable improvements in personalized 40 Hz entrainment therapy for Alzheimer’s disease?

Hypothesis: A Temporal Convolutional Network trained on 12 causal PAC-derived and stimulation context features — rather than 73 spectral features — can predict future PAC at horizons where simpler baselines fail (beyond ≈ 3 seconds), enabling a predictive controller that outperforms reactive threshold control and approaches the theoretical oracle bound.

Project Origin

I became interested in computational approaches to Alzheimer’s therapy after reading about the landmark Iaccarino et al. (2016) Nature study showing that 40 Hz sensory stimulation reduced amyloid-beta plaques in AD mouse models by up to 50%. Looking into human clinical translation, I noticed a gap: all existing protocols deliver stimulation on rigid fixed schedules that ignore individual neural responses. Approximately 30% of patients are non-responders (Fortunato et al., 2023), and habituation degrades entrainment within sessions — yet fixed protocols cannot detect or adapt to these dynamics.

This observation led to my central question: could machine learning predict when a patient’s brain will lose entrainment, enabling proactive intervention? I began this project in early 2026 using the publicly available OpenNeuro ds005048 EEG dataset. The project concept originated from my independent literature review and was not assigned as part of any class or institutional program.

Continuation

None. This is not a continuation of any previous science fair project.

Work by Others

Key prior work relevant to this project:

- **Iaccarino et al. (2016, Nature):** 40 Hz optogenetic and sensory stimulation reduced amyloid-beta by ~50% in AD mouse models through microglial activation — the foundational study for 40 Hz entrainment therapy.
- **Murdock et al. (2024, Nature):** Identified glymphatic clearance as a key mechanism; multisensory 40 Hz stimulation reduced neocortical plaque volume by 37% in 5XFAD mice.
- **Chan et al. (2025, Alzheimer's & Dementia):** Open-label extension study showing sustained EEG entrainment and reduced brain atrophy in mild AD patients with 40 Hz multisensory stimulation — first human evidence of target engagement at this scale.
- **Fortunato et al. (2023, Frontiers in Neuroscience):** Found ~30% non-responder rate in 40 Hz entrainment studies, motivating personalized approaches.
- **Lahijanian et al. (2024, Scientific Reports):** Created the OpenNeuro ds005048 dataset (35 elderly subjects, 40 Hz auditory entrainment) and showed enhanced default mode network connectivity.
- **Lawhern et al. (2018, J Neural Engineering):** Developed EEGNet, a compact CNN for brain-computer interfaces, which I adapted as the static PAC estimator.
- **Tort et al. (2010, J Neurophysiology):** Introduced the Modulation Index for measuring phase-amplitude coupling, used as the entrainment biomarker throughout this work.
- **Cabral et al. (2025, Frontiers in Digital Health):** Advocated for AI-driven biofeedback for personalized digital therapeutics in gamma entrainment.

Methods

Dataset

- OpenNeuro ds005048 (Lahijanian et al., 2024)
- 35 elderly subjects: Alzheimer’s disease (n = 17), MCI (n = 6), healthy controls (n = 10), unspecified (n = 2)
- 7 frontal EEG channels (Fp1, Fp2, F7, F3, Fz, F4, F8), sampled at 250 Hz
- Protocol: 40 Hz auditory stimulation (40 s ON / 20 s OFF cycles)

Preprocessing

- Bandpass filter: 0.5–80 Hz (4th-order Butterworth, zero-phase)
- Notch filter: 50 Hz (Q = 30) for power-line removal
- Artifact rejection: ± 100 μ V threshold (applied before CAR)
- Common average reference (after artifact rejection)
- Windowing: 2-second windows (500 samples), 1-second hop

Subject-Level Data Splits (No Within-Subject Leakage)

- Training: 24 subjects (11,736 windows)
- Validation: 5 subjects (2,725 windows)
- Test: 6 subjects (2,822 windows) — never seen during training or hyperparameter selection

Phase-Amplitude Coupling (PAC) Computation

- Modulation Index (Tort et al., 2010): measures coordination between theta phase (4–8 Hz) and gamma amplitude (38–42 Hz)
- KL divergence from uniform distribution over 18 phase bins (20° each)
- Computed at epoch level (20–40 s); assigned to all constituent 2-second windows within each epoch
- Range: 6e-6 to 7e-4 (dimensionless MI units); mean approx. 4.4e-5

Feature Engineering: PAC + Stimulation Context (12 Features)

A feature ablation study revealed that 61 spectral features encode subject-specific EEG characteristics (likely reflecting individual anatomy and recording conditions) that do not generalize across subjects. Dropping all spectral features and using only 12 PAC-derived and stimulation context features raised test R^2 from -0.025 (73 features) to 0.558 (12 features, single seed). The 5-seed mean is $R^2 = 0.606 \pm 0.032$, a $5\times$ improvement over the prior TCN best of 0.121 .

Feature Subset	# Features	Test R^2
All (spectral + PAC + stim)	73	-0.025
PAC only	7	0.344
PAC + Stim context (final)	12	0.558^*
Spectral only	61	-0.420

**Single-seed ablation result. 5-seed mean: $R^2 = 0.606 \pm 0.032$ (see Results, Table 3).*

The 12 PAC+Stim features: PAC current value, causal moving averages (2, 4, 8, 16 windows), first-order and 4-step differences, stim state (binary), time since switch, stim fraction over 20 s, cycle phase (sin + cos encoding). All strictly causal. Z-score normalized on training set only.

Stage 1: EEGNet — Static PAC Estimator

- Input: (batch, 1, 7, 500) — 7 channels, 2 seconds at 250 Hz
- Block 1: temporal convolution (8 filters, 256 ms kernel) + depthwise spatial convolution across 7 channels
- Block 2: separable convolution (16 filters); fully connected regression head outputting scalar PAC prediction
- 1,457 parameters; MSE loss; Adam optimizer; test $R^2 = 0.287$ (data-imposed ceiling; best epoch 53)

Stage 2: MultiscaleCausalTCN — Temporal Forecaster

- Input: (batch, 20, 12) — 20-step lookahead (20 seconds) $\times$ 12 PAC+Stim features
- 4 causal depthwise-separable conv blocks with dilations [1, 2, 4, 8]; effective receptive field = 31 time steps
- GroupNorm + SiLU activation; attention pooling; dual-head output (future PAC + delta-PAC)
- 22,914 parameters ($h = 64$); Huber loss; AdamW; early stopping (patience = 20 epochs)
- 5-second prediction horizon; test $R^2 = 0.606$ (7ch, 5-seed mean 0.606 ± 0.032 , range 0.558 – 0.647); 4ch test $R^2 = 0.430$ ($h = 32$, 5,154 params)

Closed-Loop Controller Design

- **Personalization:** 30-second rolling baseline per subject; $z\text{-score} = (\text{PAC} - \mu) / \sigma$
- **Decision logic:** $z < -0.5 \rightarrow \text{STIMULATE}$; $z > +0.5 \rightarrow \text{REST}$; else MAINTAIN current state
- 5-second hysteresis to prevent rapid state oscillation
- **Variants tested:** Fixed Schedule (40 s ON / 20 s OFF), Reactive Threshold, TCN Predictive, Hybrid, PI Controller, Alignment Oracle (perfect hindsight upper bound)

Validation Protocol

- **Offline counterfactual replay** on all 35 subjects' recorded EEG — not simulated brain dynamics
- Ground-truth PAC labels used as TCN input to isolate the forecaster's predictive contribution from EEGNet error
- **Metrics:** Alignment = mean of Low-PAC Stim Rate and High-PAC Rest Rate; PAC Targeting Gap (mean PAC during rest minus mean PAC during stimulation)
- **Statistics:** Wilcoxon signed-rank (paired, $N = 35$); Hedges' g with 95% BCa bootstrap confidence intervals (10,000 iterations)

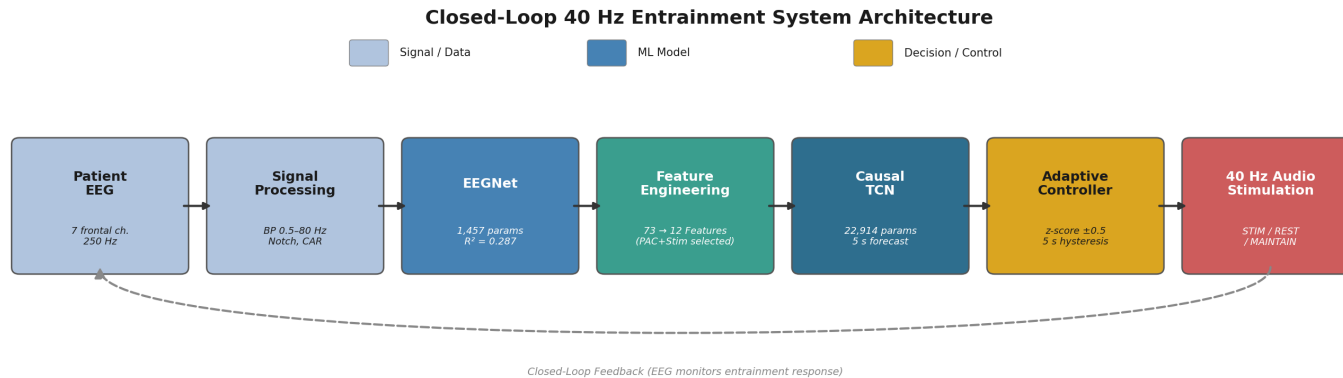


Figure 1. System architecture of the closed-loop 40 Hz entrainment system. Raw EEG from 7 frontal channels flows through signal processing, EEGNet PAC estimation, 12-feature PAC+Stim engineering, causal TCN forecasting (5 s horizon), and an adaptive controller that drives personalized 40 Hz auditory stimulation. Dashed arrow indicates closed-loop feedback. (Diagram created by the author.)

Results

Architecture Search: The $R^2 = 0.287$ Static Ceiling

Eight neural network architectures (1,457 to 1.1 M parameters) were tested for static PAC prediction from 2-second EEG snapshots. All converged to $R^2 \approx 0.287$ — a data-imposed ceiling showing that epoch-level PAC cannot be recovered from instantaneous windows. This finding redirected the project toward temporal forecasting. **The bottleneck was in the features, not the architecture.**

Horizon Sweep: PAC+Stim TCN vs. Baselines

TCN models trained on 12 PAC+Stim features at horizons 1–10 s (PAC+Stim features only). An inflection point emerges at ≈ 3 s where persistence collapses but the TCN maintains strong R^2 :

Horizon	Persistence R^2	TCN R^2 (7ch)	TCN R^2 (4ch)	TCN Margin
1 s	0.726	0.725	0.642	−0.001
3 s	0.178	0.607	0.391	+0.429
5 s	0.104	0.577	0.398	+0.473
8 s	−0.007	0.370	0.419	+0.377
10 s	−0.081	0.669	0.387	+0.750

Table 1. PAC+Stim TCN horizon sweep (single seed). At 1 s, persistence is competitive. At 3–10 s, persistence collapses while the TCN (7ch) maintains $R^2 = 0.37$ – 0.67 — the operationally actionable regime for proactive neuromodulation. Multi-seed 5-seed mean at horizon 5: $R^2 = 0.606 \pm 0.032$, range 0.558–0.647.

Controller Comparison (N = 35 Subjects, Real EEG)

Controller	Alignment	Low-PAC Stim	Stim %	PAC Gap
Fixed Schedule	45.0%	61.4%	66.6%	−6.6
Reactive Threshold	64.5%	51.7%	36.7%	+21.1
TCN Predictive	72.1%	82.6%	59.7%	+30.5
Alignment Oracle	100%	100%	48.3%	+33.3

Table 2. Controller comparison on 35 subjects' real EEG recordings. PAC Gap in dimensionless Modulation Index units ($\times 10^{-6}$ MI units). The TCN Predictive controller reaches 91% of the theoretical oracle bound.

Multi-Seed Robustness (7ch, horizon = 5 s)

Validated across 5 random seeds (h = 64 TCN, PAC+Stim features):

Seed	Val R^2	Test R^2
42	0.804	0.558
123	0.822	0.620
456	0.799	0.597
789	0.831	0.608
2024	0.846	0.647
Mean $\pm$ Std	0.820 $\pm$ 0.019	0.606 $\pm$ 0.032

Statistical Significance (TCN vs. Reactive Threshold)

- **Alignment:** 72.1% vs. 64.5% — Hedges' $g = +1.31$ [95% CI: 0.75, 1.87], $p < 0.001$ (large effect)
- **Low-PAC Stim Rate:** 82.6% vs. 51.7% — Hedges' $g = +4.47$ [95% CI: 3.33, 5.62], $p < 0.001$ (very large effect)
- **PAC Targeting Gap:** 30.5 vs. 21.1 ($\times 10^{-6}$ MI) — Hedges' $g = +1.57$ [95% CI: 0.98, 2.17], $p < 0.001$ (large effect)
- **Universal benefit:** 35/35 subjects showed higher alignment with TCN control (binomial $p < 0.001$); advantage consistent across train (24), validation (5), and held-out test (6) splits.
- **Threshold robustness:** TCN outperforms reactive at all z-score thresholds from 0.2 to 1.0; stable performance plateau at ≥ 0.3 .

Discussion

Interpretation of Results

- **Feature selection matters more than architecture:** 8 static model families all converged to $R^2 \approx 0.287$ on 73 features; switching to 12 PAC+Stim features raised test R^2 to 0.606. A post-hoc comparison of 10 temporal architectures (TCN, Transformer, GRU, LSTM, CNN, XGBoost, Ridge, and linear models) confirmed that all Tier 1 architectures converge to $R^2 \approx 0.61$ – 0.65 on the same 12 features, while none exceeds $R^2 = 0.28$ on 73 features. The bottleneck was always in the input representation, not the model.
- The ≈ 3 -second inflection point reflects PAC autocorrelation timescale: below 3 s, persistence suffices; above 3 s, state transitions require temporal context that only the TCN captures from the 20-step PAC trajectory.
- Spectral features encode subject-specific EEG characteristics (likely individual anatomy and recording conditions) that don't generalize: val-test R^2 gap shrinks from 0.358 (73 feat) to 0.246 (12 feat) when spectral features are removed.
- The 60% improvement in low-PAC targeting (82.6% vs. 51.7%) matters most clinically: the controller concentrates therapy where neural coupling is weakest.

Comparison to Prior Work

- Analogous to closed-loop DBS for Parkinson's disease (Rosin et al., 2011): adaptive stimulation concentrates therapy on periods of genuine need.
- Extends beyond spindle detection (Portiloop, Lacroix et al., 2022) to multi-step forecasting — necessary when proactive intervention requires advance lead time.
- First system targeting PAC dynamics specifically for 40 Hz gamma entrainment optimization at 5–10 s horizons.

Limitations and Possible Errors

- **Offline counterfactual replay**, not live closed-loop: measures decision quality, not realized therapeutic benefit. Live validation with real-time EEG streaming is required.
- **EEGNet not in the validation loop:** ground-truth PAC was used as TCN input; a deployed system would propagate EEGNet estimation error ($R^2 = 0.287$) into forecasts.
- **Single-site dataset** (Tehran memory clinic): generalizability to other populations, stimulation modalities, and EEG equipment is unknown.
- **7 frontal channels only:** relevant theta–gamma coupling in parietal and temporal regions is not captured by this

Conclusions

Key Findings

- Eight architectures (1,457 to 1.1 M parameters) all converge to $R^2 = 0.287$ for static PAC prediction — a data-imposed ceiling proving the bottleneck is the feature representation, not the model.
- **Feature discovery:** replacing 73 spectral features with 12 PAC+Stim features raised temporal test R^2 from -0.025 to 0.606 (5-seed mean; 4ch: 0.430). Feature selection mattered more than any architectural change.
- The TCN maintains $R^2 = 0.37$ – 0.67 at 3–10 s horizons where persistence collapses to negative R^2 , defining the operationally actionable regime for proactive neuromodulation.
- The predictive controller achieves 72.1% alignment, 82.6% low-PAC targeting, and 91% of the theoretical oracle bound (all $p < 0.001$ vs. reactive; Hedges' $g = 1.31$ – 4.47).
- All 35 subjects benefited, including 6 held-out test subjects never seen during training.

Context

These results show that temporal PAC forecasting can drive proactive closed-loop control that outperforms both fixed-schedule and reactive protocols. No prior system has attempted PAC-specific temporal prediction for personalized gamma entrainment control. This work is a computational validation; live closed-loop trials are needed to confirm clinical translation.

Productization and Clinical Roadmap

- **Caregiver web application:** deployed on Hugging Face Spaces (Streamlit); PAC monitoring with simulated EEG, real-time stimulation control, and personalized session tracking — a prototype interface for clinician monitoring.
- **Phase 1 — Observational (planned):** collect real-world EEG + PAC data via app to characterize inter-session habituation patterns. No intervention arm.
- **Phase 2 — Feasibility (12–18 months):** IRB-approved within-subject pilot ($N = 5$ – 10 healthy adults) comparing adaptive TCN vs. fixed-schedule 40 Hz stimulation, verifying alignment improvement and real-time latency.
- **Phase 3 — Comparative (3–5 years):** randomized controlled trial in mild AD patients ($N \approx 20$) comparing personalized adaptive vs. fixed protocol. Primary endpoint: maintained entrainment at 12-week follow-up.
- **Near-term technical:** end-to-end validation (raw EEG $\rightarrow$ EEGNet $\rightarrow$ TCN $\rightarrow$ controller); per-subject online fine tuning; 4 channel Muse compatible deployment.

Scope of Work

New Work by Author

- Independently identified the research gap (fixed-schedule limitation in 40 Hz entrainment) through literature review.
- Designed and implemented the full computational pipeline: EEG preprocessing, PAC computation, architecture search across 8 neural network families, feature ablation study ($73 \rightarrow 12$ PAC+Stim features), and MultiscaleCausalTCN design.
- Discovered the $R^2 = 0.287$ data ceiling through experimentation, the ≈ 3 -second prediction horizon inflection point, and the feature selection breakthrough (12 PAC+Stim features raising R^2 from -0.025 to 0.606) — all novel empirical findings.
- Designed the closed-loop controller with personalization module, hysteresis logic, and multi-strategy comparison framework including alignment oracle upper bound.
- Conducted all statistical analyses: Wilcoxon signed-rank tests, Hedges' g effect sizes with 95% bootstrap CIs, binomial sign tests, threshold sensitivity sweeps.
- Validated the complete system on 35 real patient EEG recordings through offline counterfactual replay.

Professional, Institutional, and Academic Resources and Support

- **Data:** OpenNeuro ds005048 — publicly available EEG dataset accessible to anyone at no cost.
- **Computing:** personal Apple Silicon MacBook; no institutional computing resources, GPU clusters, or specialized laboratory equipment were used.
- **Software:** open-source libraries (PyTorch, MNE-Python, SciPy, NumPy, Matplotlib) — all freely available.
- No institutional lab, university mentor, summer research program, or specialized equipment was used at any stage.

References / Supplemental Information

References

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Supplemental Information

Dataset: <https://openneuro.org/datasets/ds005048>